UNBench Reproduction and Optimization Report

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Project Summary & Reproduction Goal

Project Summary: UNBench

UNBench evaluates AI understanding of global governance by simulating United Nations Security Council (UNSC) tasks, specifically focusing on geopolitical reasoning capabilities.

Goal: Co-penholder Judgment

Reproduce Task 1 where an LLM must identify the correct co-author for a resolution draft from a list of 2 to 5 choices, testing geopolitical inference accuracy.

Setup Notes

Environment

Python-based environment using `openai` and `tqdm` libraries.

Data Source

Utilized the `task1.json` dataset and resolution text files from the `data/task1` directory.

API Configuration

Accessing models via the `aihubmix.com` endpoint using a DeepSeek API key.

Compute Infrastructure

Executed via a Google Colab notebook environment for computational resources.

Reproduction Target & Metric

Reproduction Target

Our goal is to achieve high accuracy in predicting the ground-truth co-author across multiple difficulty tiers.

Difficulty Tiers:

- 2 Choices
- 3 Choices
- 4 Choices
- 5 Choices

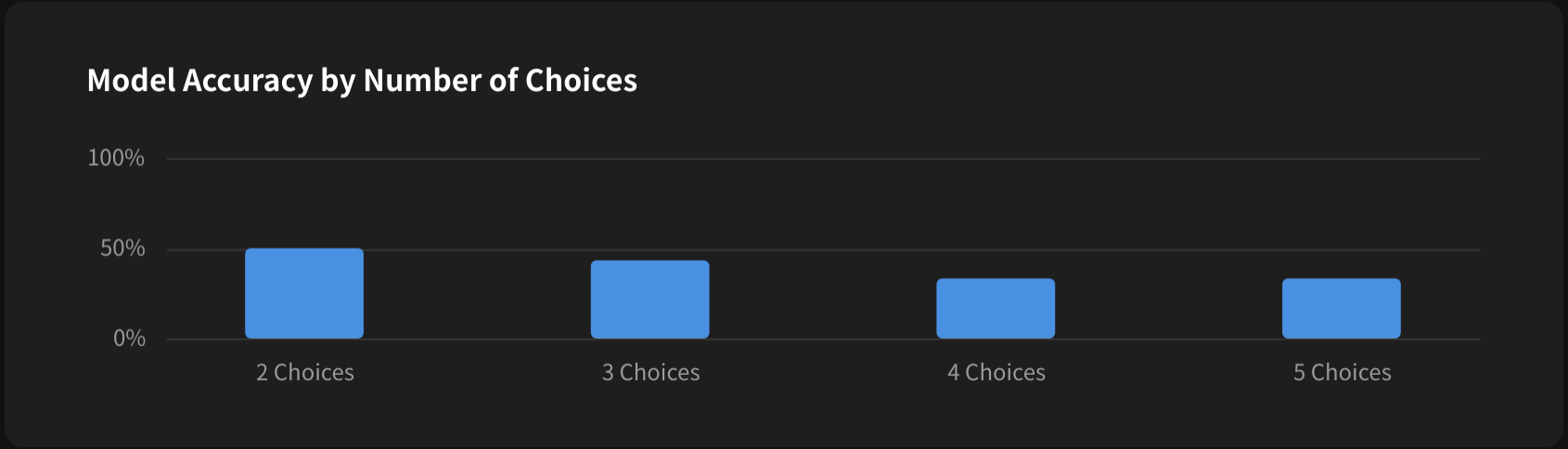
Evaluation Metric

We use the standard Accuracy metric to measure performance.

Formula Definition:
Accuracy = (Number of Correct Matches) / (Total Predictions)

In code terms:correct / len(results)

Initial Results & Analysis



Key Observation

Baseline results showed a significant drop in accuracy as the number of options increased, particularly in the 5-choice scenario, indicating limitations in complex reasoning.

Modification & Optimized Results

Key Modifications (Mod4)

Generation Strategy Tuning

- Temperature = 0.1 (Higher Precision)
- Top-p = 0.9 (Better Decision Quality)

Penalty Enforcement

- Frequency Penalty = 0.2
- Presence Penalty = 0.1 (Focus on Country List)

Performance Breakthrough

5-Choice Accuracy (vs Baseline)

0.333
Before Optimization

0.500
After Optimization

Substantial Improvement

The accuracy has seen a significant jump, demonstrating the effectiveness of the modified generation strategy and penalty mechanism in focusing the model's attention.

Debug Diary

Blocker 1: Invalid Responses

Problem: Unwanted "Chatty" Output. The model provided extra text or explanatory sentences, causing failures in strict string-matching tests.
Resolution: Constrained Prompting. Implemented strict system prompts and reduced `max_tokens` to 15 to enforce single-word output.

Blocker 2: Deterministic Bias

Problem: At `temperature=0.0`, the model occasionally made mechanical, incorrect choices consistently.
Resolution: Introduced a slight fluid variance (`temperature=0.1`) and top_p sampling to allow the model to select more "strategically" reasonable options.

Conclusions

Highly Reproducible

The basic task structure and scoring mechanism of UNBench are confirmed to be highly stable and reproducible.

Limitations (No Tuning)

High accuracy in complex 5-choice scenarios is not easily achieved using default "out-of-the-box" settings.

Root Cause Analysis

Geopolitical reasoning requires fine-tuned hyperparameters to prevent hallucinations and ensure strict adherence to formatting constraints.

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